Computer Organization
Homework 2
(Due on 2023/11/2)

Note: Please hand in the homework on time.

1. For the following C statement, what is the corresponding MIPS assembly code? Assume that the variables x , y , and z are saved in $s0, $s1, $s2,$s3. Please include instruction “addi”. (10%)

```
y= ( x+5 ) − (z−10);
```

2. For the following MIPS assembly instructions above, what is a corresponding C statement? (10%)

```
sub  a, b, c;
sll   d, a, 5;
```

3. Assume that the variables i and j are assigned to register $s2, $s3. Assume that the base address of arrays P and Q are in register $s6 and $s7, respectively. Variable ACC are assigned to register $s5. For the MIPS assembly instruction below, what is the corresponding C statement? Please answer (a) to (e) step by step. (20%)

sll	$t0, $s2, 2	
add	$t0, $s6, $t0	
sw	$s5, 8($t0)	# (a) What is the meaning of these three lines (Please use the variables in the statement of the question if the registers are defined.)
sll	$t2, $s3, 2	
add	$t2, $s7, $t2	
lw	$t0, 0($t2)	# (b) What is the meaning of these three lines
add	$s5, $s5, $t0	# (c) What is the meaning of this line?
lw	$t0, -4($t2)	# (d) What is the meaning of this line?
sub	$s5, $s5, $t0	# (e) What is the meaning of this line?

4. Translate the following C code to MIPS. Assume that the variables j is assigned to register $s1. Assume that the base address of arrays A and B are in register $s2 and $s3, respectively.(20%)

$$A[j] = B[2j] - B[2j-3]$$

5. Assume that registers $s0 and $s1 hold the values 0x7FFFFFFF and 0x90000002, respectively.

(a) What is the value of $t7 for the following assembly code? Please express in hexadecimal. (5%)

```
sub $t7, $s0, $s1
```

(b) What is the result in $s2 and $s3 for the following instruction? (10%)

```
slti $s2, $s1, -1
sltu $s3, $s0, $s1
```

6. Consider the following MIPS loop:

```
LOOP: slt  $t2, $0,   $t1
      beq  $t2, $0,  DONE
      subi $t1, $t1, 1
      addi $s2, $s2, 5
      j    LOOP
DONE:
```

(a) Assume that the register $t1 is initialized to the value 15. What is the value in register $s2 assuming $s2 is initially zero? (5%)

(b) For each of the loops above, write the equivalent C code routine. Assume that the registers $s1, $s2,and $t1are integers A, B, and i, respectively. (10%) (hint:Use while loop or for loop.)

(c) For the loops written in MIPS assembly above, assume that the register $t1 is initialized to the value N. How many MIPS instructions are executed? (10%)